SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING COMPUTER SCIENCE AND ENGINEERING LIST OF EXPERIMENTS

SUB CODE & NAME: CSA1712-ARTIFICIAL INTELLIGENCE

5.Write the python program for Missionaries Cannibal problem

Code:

```python
#pg 5
from collections import deque

start = (3, 3, 1)
goal = (0, 0, 0)

queue = deque([(start, [])])
visited = {start}


def safe(m, c):

    if m < 0 or c < 0 or m > 3 or c > 3:
        return False

    if m != 0 and m < c:
        return False

    rm = 3 - m
    rc = 3 - c

    if rm != 0 and rm < rc:
        return False

    return True


while queue:

    state, path = queue.popleft()

    if state == goal:
        print("Solution:")
        for s in path + [state]:
            print(s)
        break

    m, c, boat = state

    moves = [
        (1, 0),
        (2, 0),
        (0, 1),
        (0, 2),
        (1, 1)
    ]
```

.

```
for dm, dc in moves:

    if boat == 1:
        newstate = (m - dm, c - dc, 0)
    else:
        newstate = (m + dm, c + dc, 1)

    if safe(newstate[0], newstate[1]):

        if newstate not in visited:

            visited.add(newstate)

            queue.append(
                (newstate, path + [state])
            )
```

Output:

```
>>>
== RESTART: C:/Users/Lekha/AppData/Local/Programs
Solution:
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
```